
You may use one 3" x 5" note card written by you on one side only. No other materials are allowed.

1. Which is an advantage of a database over multiple flat files?
 - A. Increased data redundancy
 - B. More typing required
 - C. Better data consistency and reduced duplication**
 - D. Fewer relationships between data
2. Which situation is **most appropriate** for a non-relational database?
 - A. A college registrar storing students, majors, and courses with well-defined relationships
 - B. A bank tracking customers and accounts with strict consistency requirements
 - C. A social media application storing varying user profiles and posts whose structure changes frequently**
 - D. An airline maintaining flight reservations
3. In database terminology, an entity is:
 - A. A column in a table
 - B. A relationship between tables
 - C. An object about which information is stored**
 - D. A query result
4. Which of the following violates First Normal Form (1NF)?
 - A. One row per customer
 - B. A unique primary key
 - C. A field containing "Math, English, History" as multiple values**
 - D. One value per field
5. Consider the table below whose primary key is EmployeeID. Which normalization rule is violated?

EmployeeID	DepartmentID	DepartmentName
1001	10	Accounting
1002	20	Marketing
1003	10	Accounting

- A. First Normal Form (1NF)
- B. Second Normal Form (2NF)
- C. Third Normal Form (3NF)**
- D. None of the above

6. Consider the table below whose primary key is the combination of **StudentID** and **CourseID**. Which normalization rule is violated?

StudentID	CourseID	StudentName
101	MAT141	Smith
101	CSC150	Smith
102	MAT141	Jones

- A. First Normal Form (1NF)
- B. Second Normal Form (2NF)**
- C. Third Normal Form (3NF)
- D. None of the above

7. Consider the Access snapshot below. The fields' meanings are Last = last name, First = first name, Class Yr = class year, and Co = company. Which statement is correct?

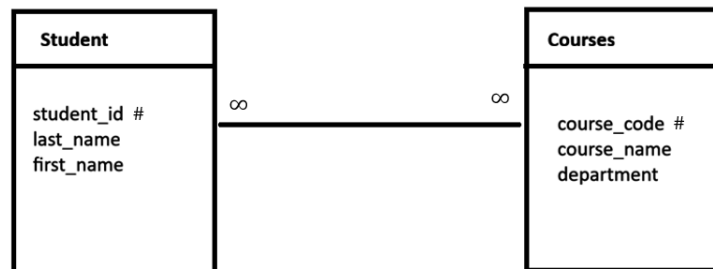
Field:	Last	First	Class Yr	Co
Table:	ORDA_Majors	ORDA_Majors	ORDA_Majors	ORDA_Majors
Sort:				
Show:	☒	☒	☒	☐
Criteria:			< 2026	= 'A'
or:				

- A. The query displays all fields from the table for records in Alpha company.
- B. The query displays only the fields Last, First, Class Yr, and Co for records with Class Yr less than 2026 and whose company is Alpha.
- C. The query displays only the fields Last, First, and Class Yr for records with Class Yr less than 2026 or whose company is Alpha.
- D. The query displays only the fields Last, First, and Class Yr for records with Class Yr less than 2026 and whose company is Alpha.

Free Response Questions

8. Sketch an ER relationship diagram for the following situation. Follow the format covered in class. Include at least one field for each entity.

One student may enroll in many courses, and one course may contain many students.



9. Consider the relationship: Department — Employee

What is the cardinality? Justify your answer, stating any assumptions you make.

Many-to-one. A department can have many employees but an employee typically belongs to a single department. (It is possible though much less likely for an employee to belong to more than one department. In this case the cardinality would be **many-to-many**.)

10. Suppose a many-to-many relationship exists between two entities in a database:

WIDGETS — WIZARDS

State whether this can be implemented directly in a relational database, or if not, how it should be addressed.

No, this cannot be implemented directly in a relational database. To address it, create a new table named WIDGETS.WIZARDS that has a compound primary key consisting of the primary keys from the WIDGETS and WIZARDS tables.